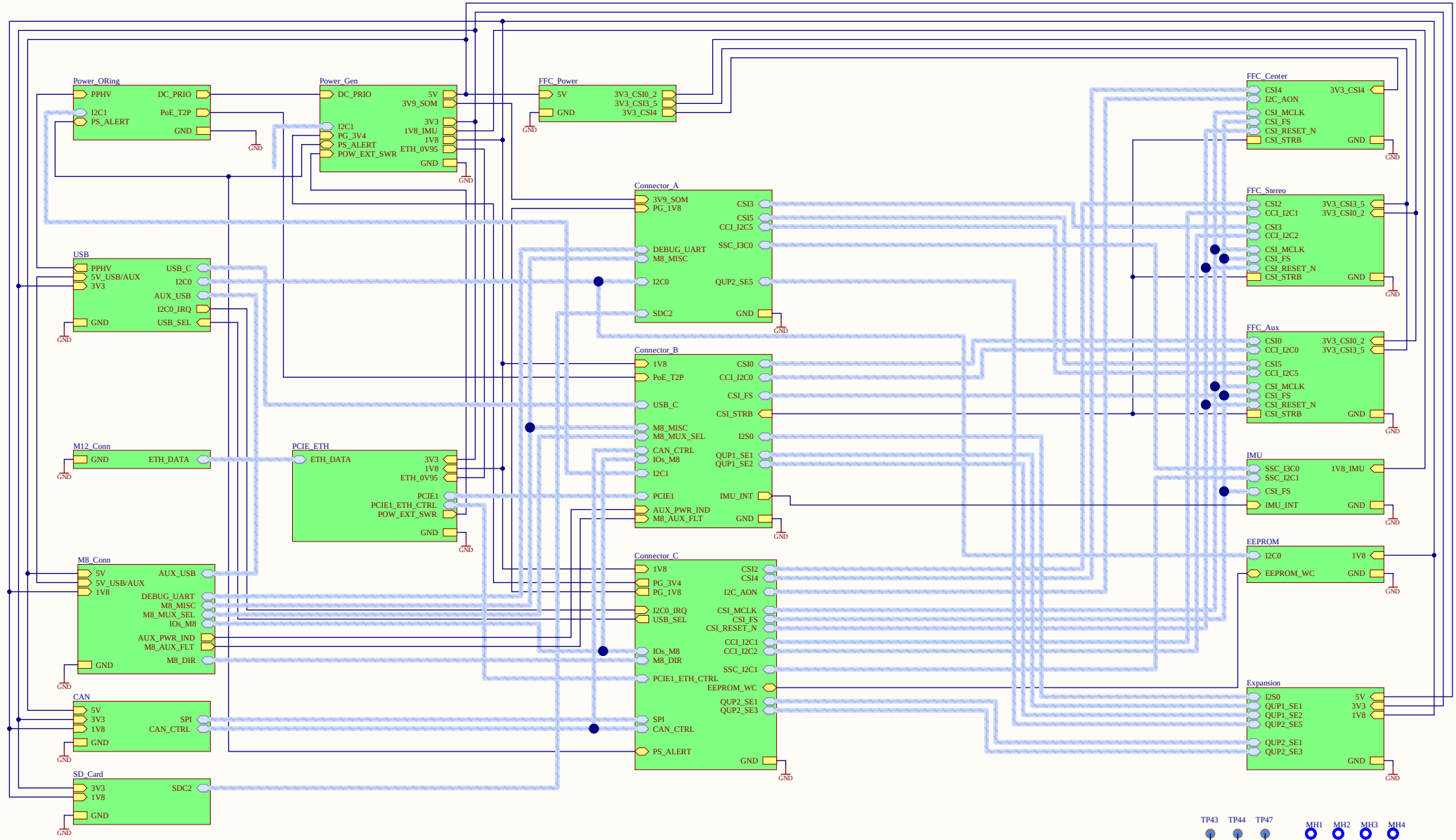
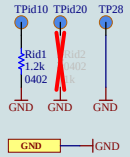

Power_ORing
PPHV
I2C1
PS_ALERT
DC_PRIO
PoE_T2P
GND
Power_Gen
DC_PRIO
I2C1
PG_3V4
PS_ALERT
POW_EXT_SWR
5V
3V9_SOM
3V3
1V8_IMU
1V8
ETH_0V95
GND
FFC_Power
5V
GND
3V3_CSI0_2
3V3_CSI3_5
3V3_CSI4
FFC_Center
CSI4
I2C_AON
CSI_MCLK
CSI_FS
CSI_RESET_N
CSI_STRB
3V3_CSI4
GND
Connector_A
3V9_SOM
PG_1V8
DEBUG_UART
M8_MISC
I2C0
SDC2
CSI3
CSI5
CCI_I2C5
SSC_I3C0
QUP2_SE5
GND
FFC_Stereo
CSI2
CCI_I2C1
CSI3
CCI_I2C2
CSI_MCLK
CSI_FS
CSI_RESET_N
CSI_STRB
3V3_CSI3_5
3V3_CSI0_2
GND
USB
PPHV
5V_USB/AUX
3V3
GND
USB_C
I2C0
AUX_USB
I2C0_IRQ
USB_SEL
Connector_B
1V8
PoE_T2P
USB_C
M8_MISC
M8_MUX_SEL
CAN_CTRL
IOs_M8
I2C1
PCIE1
AUX_PWR_IND
M8_AUX_FLT
CSI0
CCI_I2C0
CSI_FS
CSI_STRB
I2S0
QUP1_SE1
QUP1_SE2
IMU_INT
GND
FFC_Aux
CSI0
CCI_I2C0
CSI5
CCI_I2C5
CSI_MCLK
CSI_FS
CSI_RESET_N
CSI_STRB
3V3_CSI0_2
3V3_CSI3_5
GND
M12_Conn
GND
ETH_DATA
PCIE_ETH
ETH_DATA
3V3
1V8
ETH_0V95
PCIE1
PCIE1_ETH_CTRL
POW_EXT_SWR
GND
IMU
SSC_I3C0
SSC_I2C1
CSI_FS
IMU_INT
1V8_IMU
GND
EEPROM
I2C0
EEPROM_WC
1V8
GND
M8_Conn
5V
5V_USB/AUX
1V8
AUX_USB
DEBUG_UART
M8_MISC
M8_MUX_SEL
IOs_M8
AUX_PWR_IND
M8_AUX_FLT
M8_DIR
GND
Connector_C
1V8
PG_3V4
PG_1V8
I2C0_IRQ
USB_SEL
IOs_M8
M8_DIR
PCIE1_ETH_CTRL
SPI
CAN_CTRL
PS_ALERT
CSI2
CSI4
I2C_AON
CSI_MCLK
CSI_FS
CSI_RESET_N
CCI_I2C1
CCI_I2C2
SSC_I2C1
EEPROM_WC
QUP2_SE1
QUP2_SE3
GND
Expansion
I2S0
QUP1_SE1
QUP1_SE2
QUP2_SE5
QUP2_SE1
QUP2_SE3
5V
3V3
1V8
GND
CAN
5V
3V3
1V8
GND
SPI
CAN_CTRL
SD_Card
3V3
1V8
GND
SDC2
TP43
TP44
TP47
MH1
MH2
MH3
MH4
PDN_Block
Project_Information
Title BK3498
Size: Tabloid
Number:D2088000
Revision:
Date: 21/10/2026
Time: 11:51:42
Sheet 1 of 21
Drawn by: Blaz Kvas
Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States
Luxonis

Project: BK3498
Current Revision:

BK3498 Revision History:

Date	Revision		Reason for Change	Changes Implemented
06/03/2026	R0	D0		
07/06/2026	R1	D0	- Both INA700 on the same bus with same address. - Resistors beside INA are placed (should not be). - EN pin on ORing is connected to VCP pin instead of ANODE pin. - No LED indicators that would signal good power. - PG_3V4 from SOM connected to 3V3 buck instead of 5V. - Update M8 schematics (mux and level shifters) so that it's up to spec. - USB signals are connected incorrectly to the SOM. - CAN Silent pin was pulled high causing IC to always be in Silent mode. - USB TX lines should not have caps. - USB2 MUX OE pin should be connected to GND.	- Connected one INA address pin to GND. - DNPed resistor. - EN pin now connected to ANODE pin. - Added LED indicators to 5V and 3V9 rails. - Connected PG_3V4 to 5V EN pin. - Updated M8 circuit. - Reconnected MUX for USB 2 so that orientation is correct. - Connected SILENT pin to SOM so that it can be SW controlled. - Removed caps from USB TX and added 0R resistors just in case. - Connected OE pin to GND.

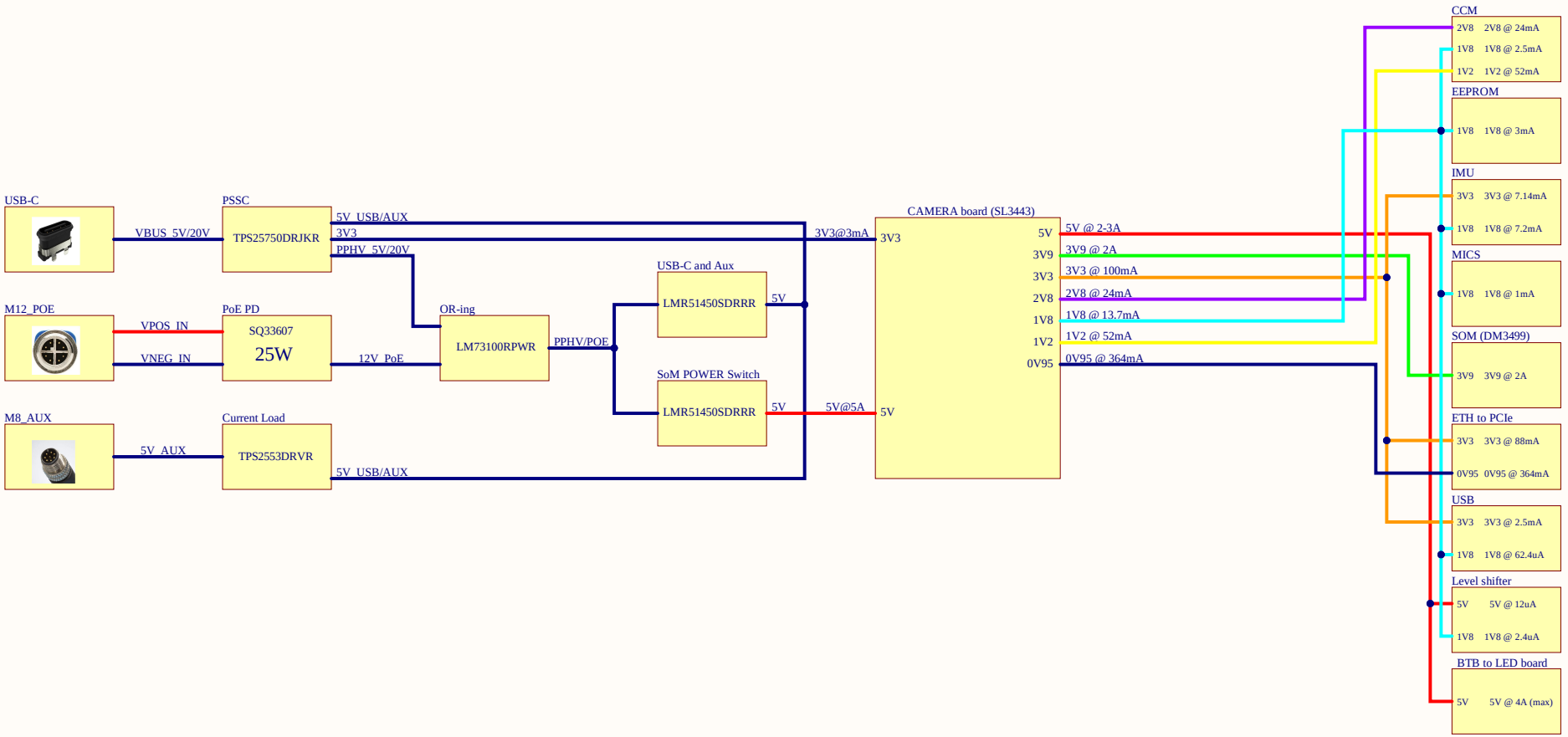


Rid1	Rid2
1K0	DNP
1K2	DNP
1K5	DNP
1K8	DNP
2k2	DNP
2k7	DNP
3k3	DNP
3k9	DNP
4k7	DNP
5k6	DNP
6k8	DNP
8k2	DNP
10k	DNP
12k	DNP
15k	DNP
18k	DNP
22k	DNP
27k	DNP
33k	DNP
39k	DNP
47k	DNP
56k	DNP
68k	DNP
82k	DNP
100k	DNP
120k	DNP
150k	DNP
180k	DNP
220k	DNP
270k	DNP
330k	DNP
390k	DNP
470k	DNP
560k	DNP
680k	DNP
820k	DNP
1k0	1k0
1k2	1k0
1k5	1k0
etc.	etc.

Rid1 / Rid2	RxDxVx
1k0 / DNP	R0D0V0
1k2 / DNP	R0D0V0

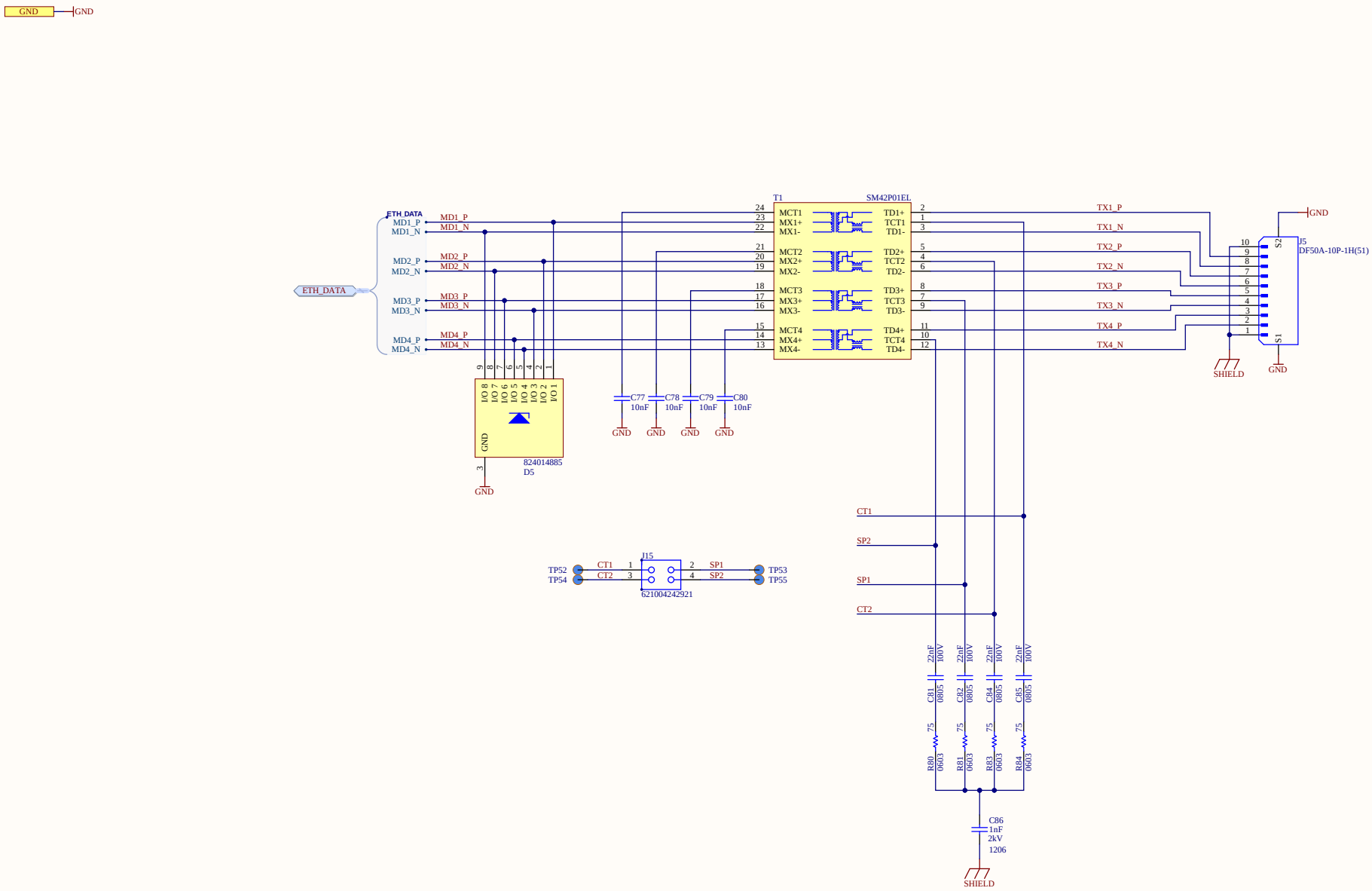
Title *BK3498*			*Luxonis Holding Corporation* *1942 Broadway St.* *STE 314C,* *Boulder, CO 80302* *United States*
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 2 of 21	
Drawn by: *Blaz Kvas*			

Power Delivery Network Block



Title **BK3498**			*Luxonis Holding Corporation* *1942 Broadway St.* *STE 314C,* *Boulder, CO 80302* *United States*
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 3 of 21	
Drawn by: ***Blaz Kvas***			

Luxonis

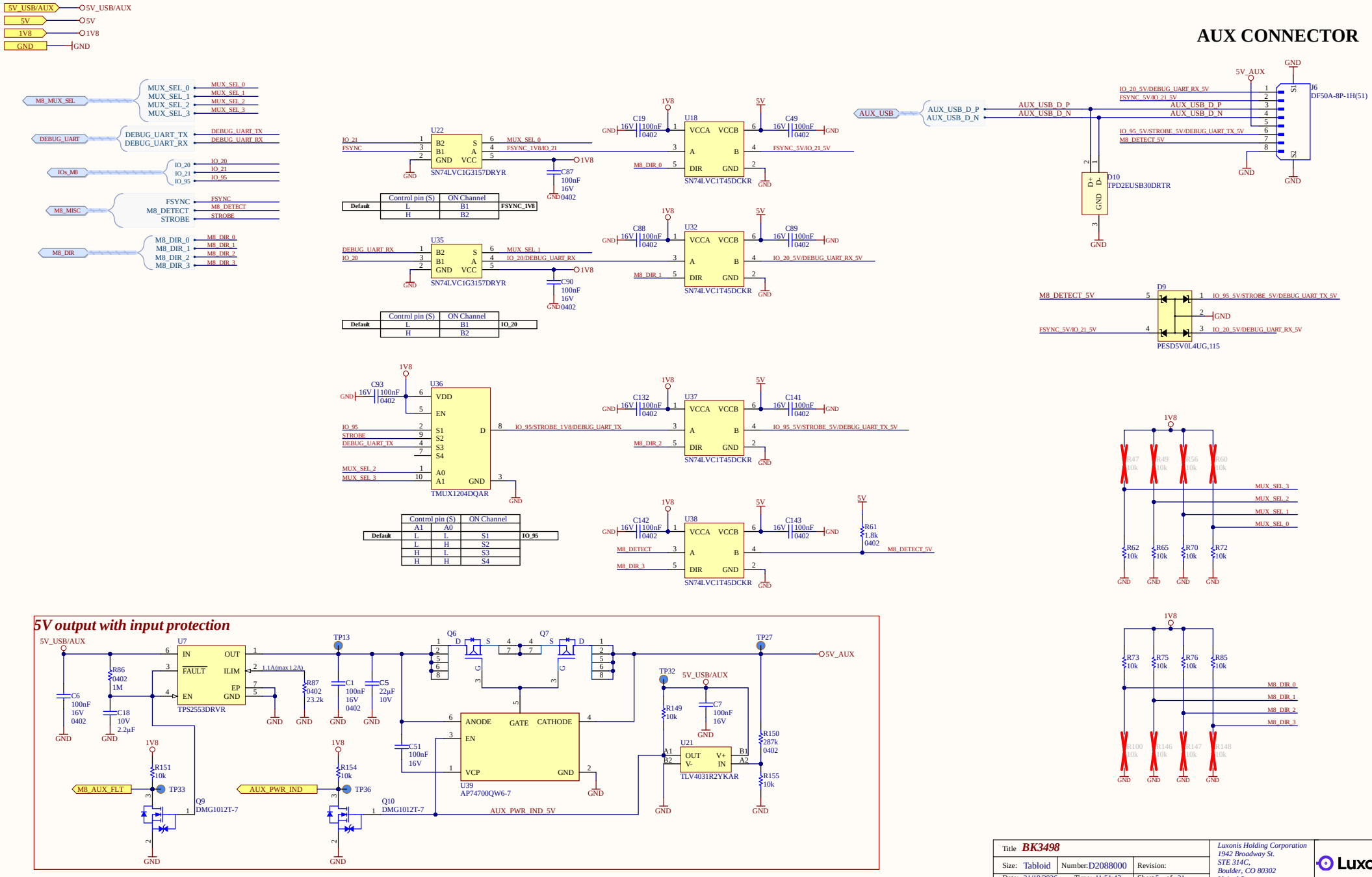


Title **BK3498**			
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 4 of 21	
Drawn by: *Blaz Kvas*			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

Luxonis

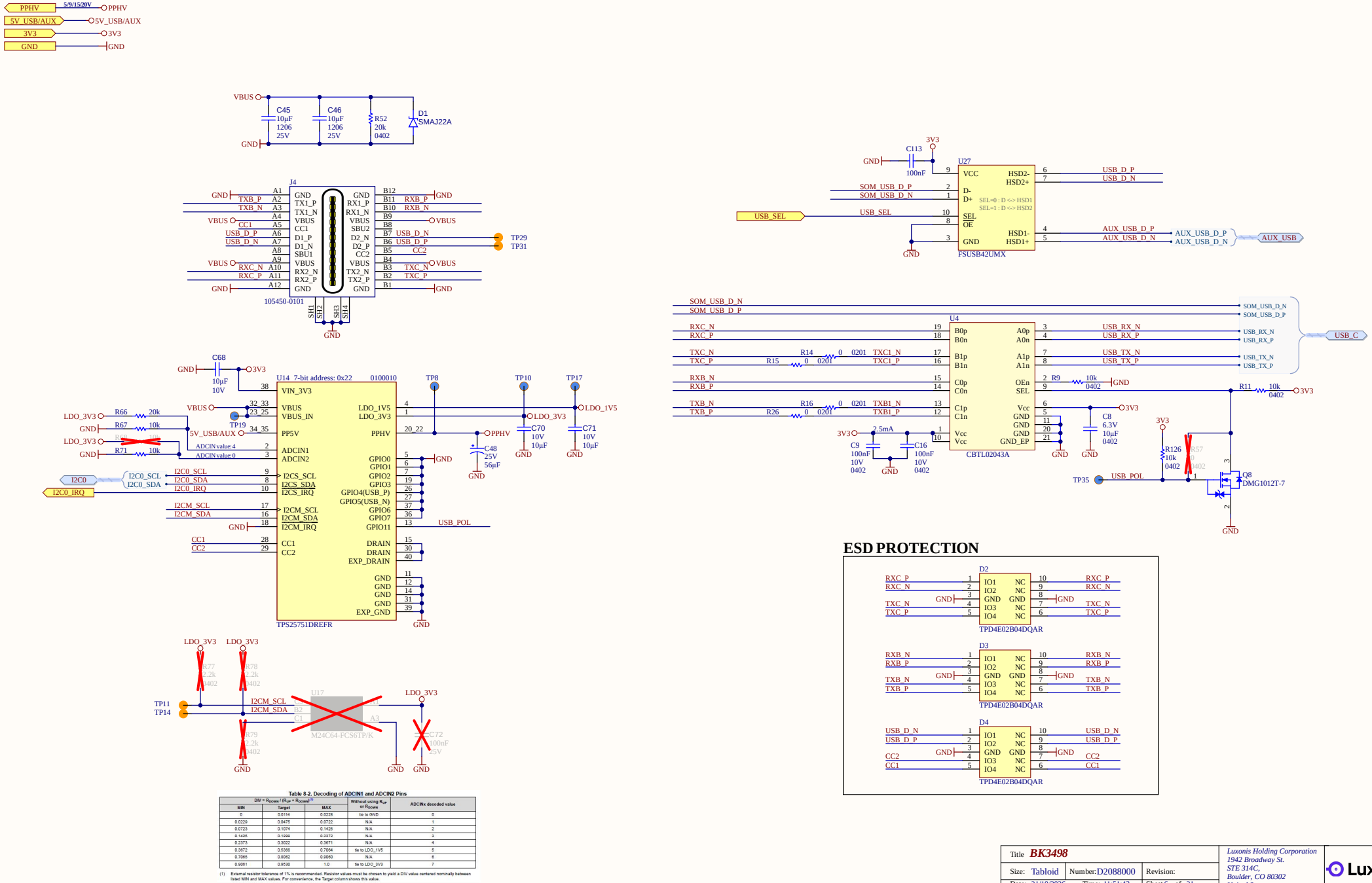
AUX CONNECTOR



Title	BK3498		
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 5 of 21	
Drawn by: Blaz Kvas			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

Luxonis



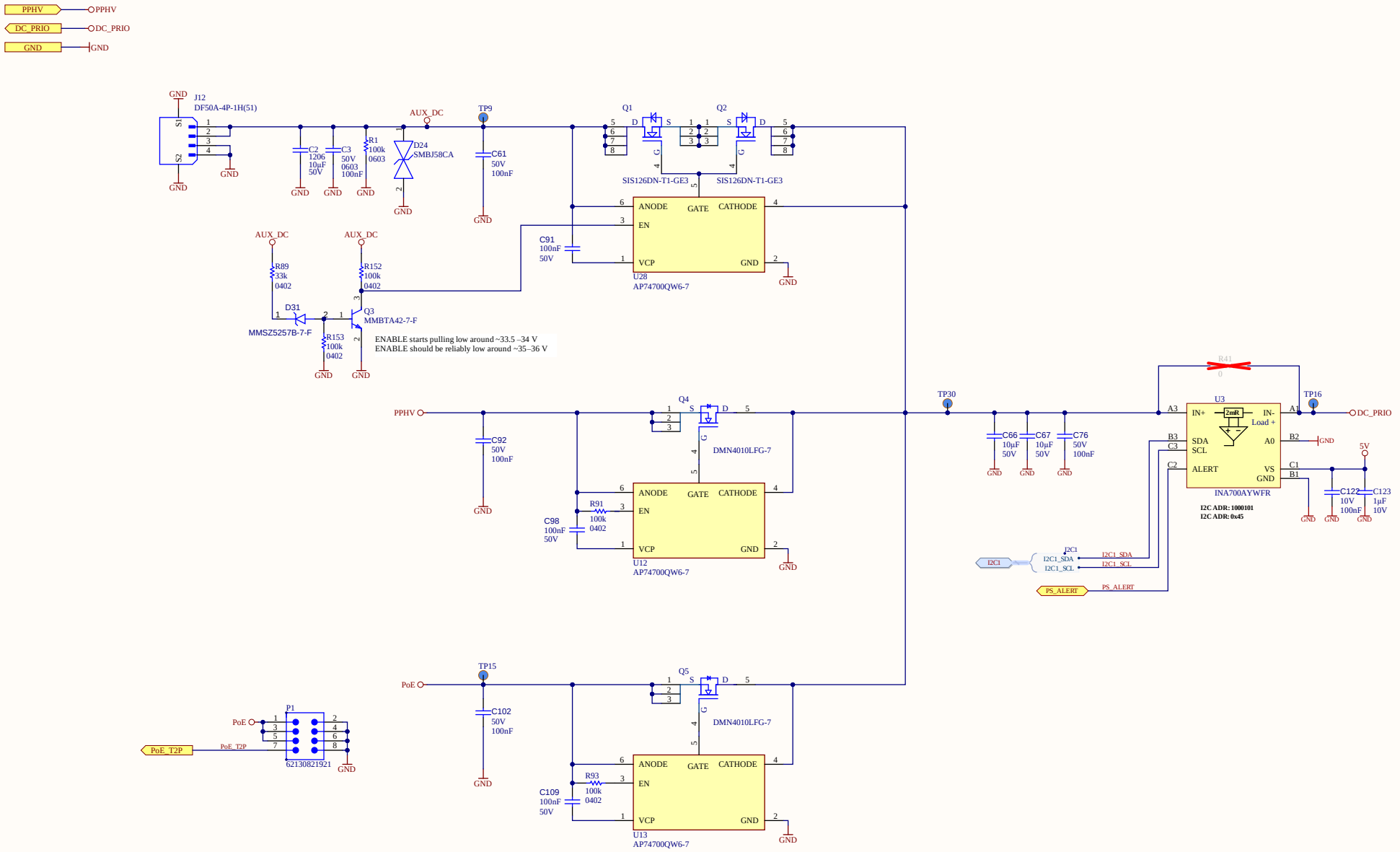
DIV = R_{DCNFx} / (R_{UP} + R_{DCNFx})[1]			Without using R_{UP} or R_{DCNFx}	ADCINx decoded value
MIN	Target	MAX		
0	0.0114	0.0228	tie to GND	0
0.0229	0.0475	0.0722	N/A	1
0.0723	0.1074	0.1425	N/A	2
0.1426	0.1999	0.2972	N/A	3
0.2373	0.3022	0.3671	N/A	4
0.3672	0.5368	0.7064	tie to LDO_1V5	5
0.7065	0.8062	0.9060	N/A	6
0.9061	0.9530	1.0	tie to LDO_3V3	7

(1) External resistor tolerance of 1% is recommended. Resistor values must be chosen to yield a DIV value centered nominally between listed MIN and MAX values. For convenience, the Target column shows this value.

Title	BK3498		
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 6 of 21	
Drawn by: Blaz Kvas			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

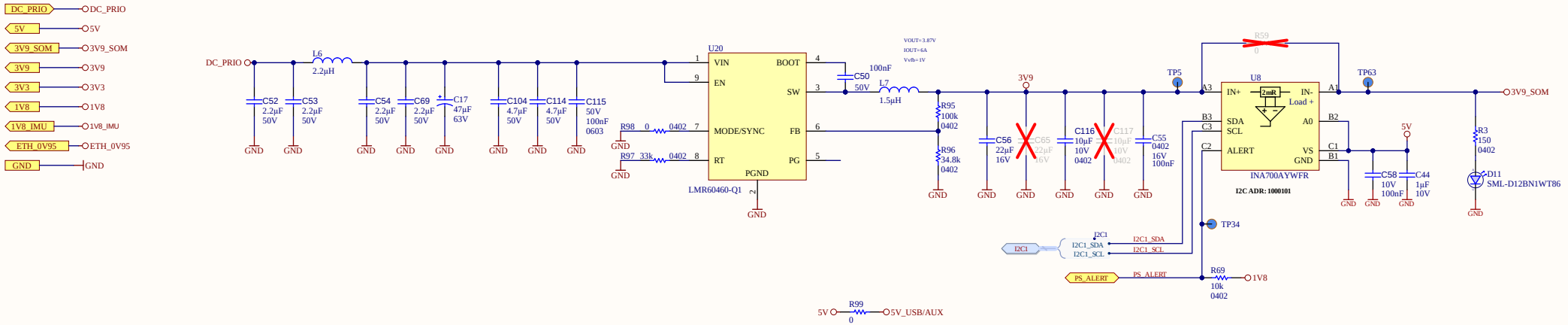
Luxonis



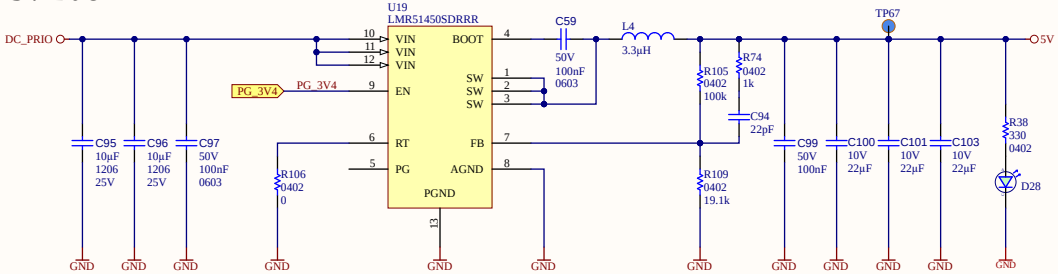
Title	BK3498		
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 7 of 21	
Drawn by: Blaz Kvas			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

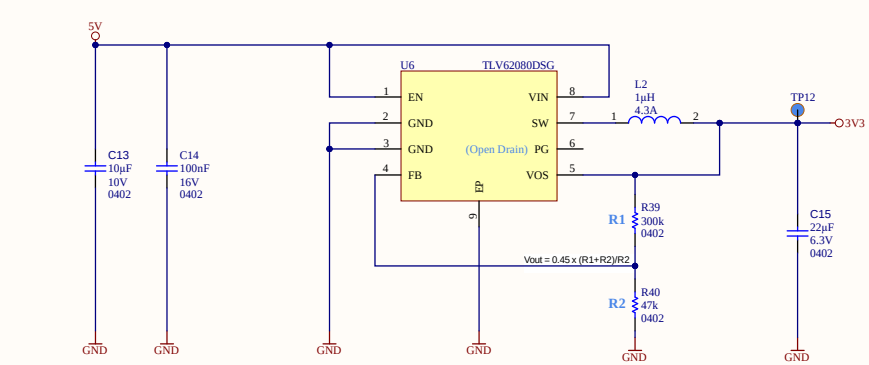
Luxonis



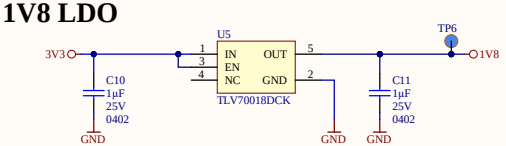
5V Buck



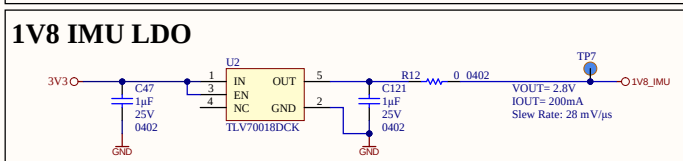
3V3 Buck



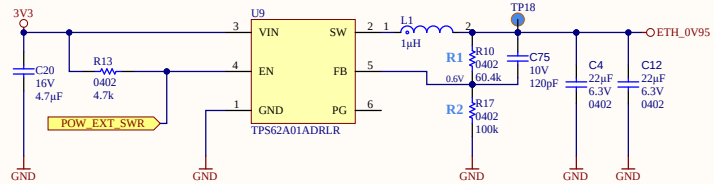
1V8 LDO



1V8 IMU LDO



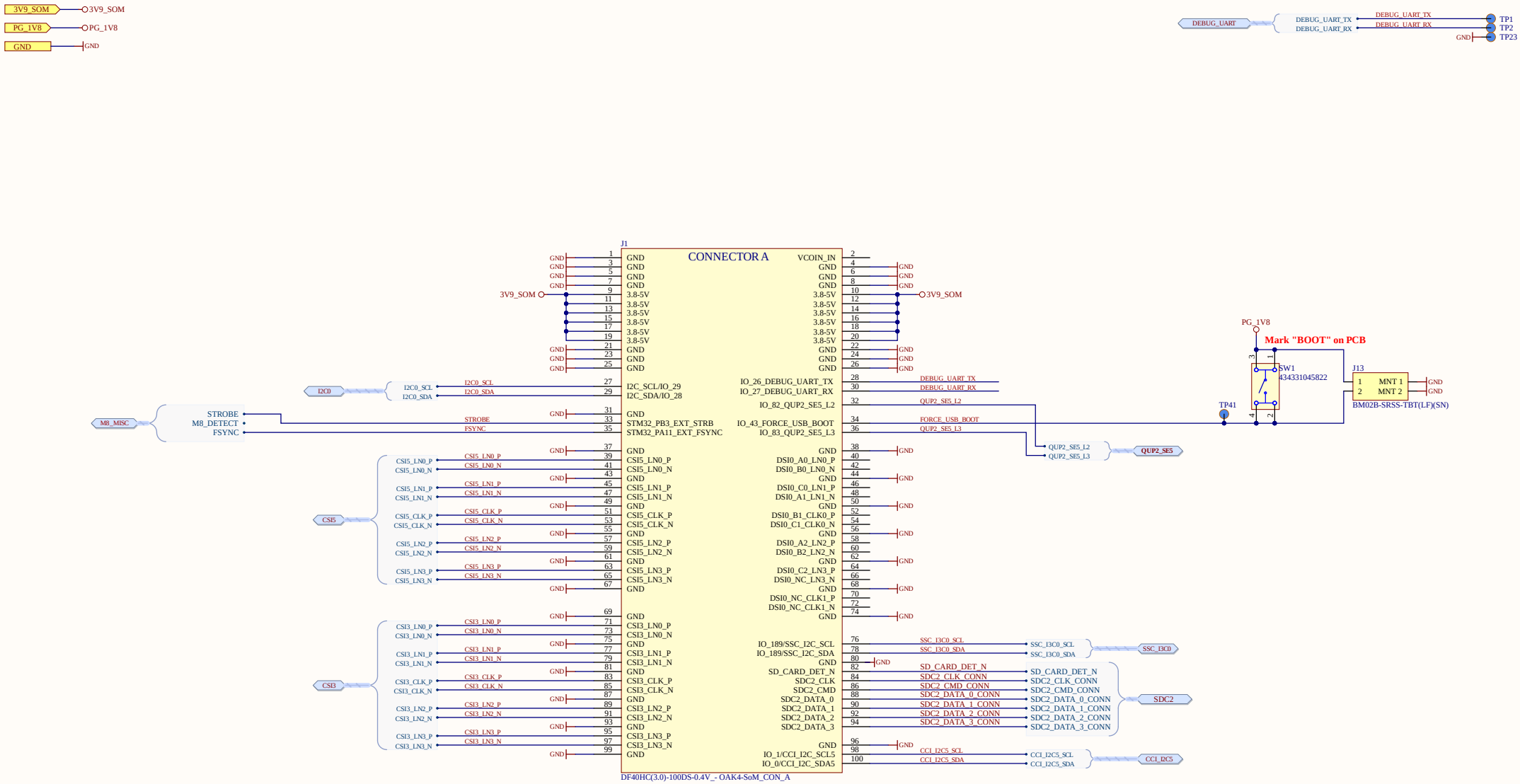
0V9 for ETH_PHY



Title	BK3498		
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 8 of 21	
Drawn by: Blaz Kvas			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

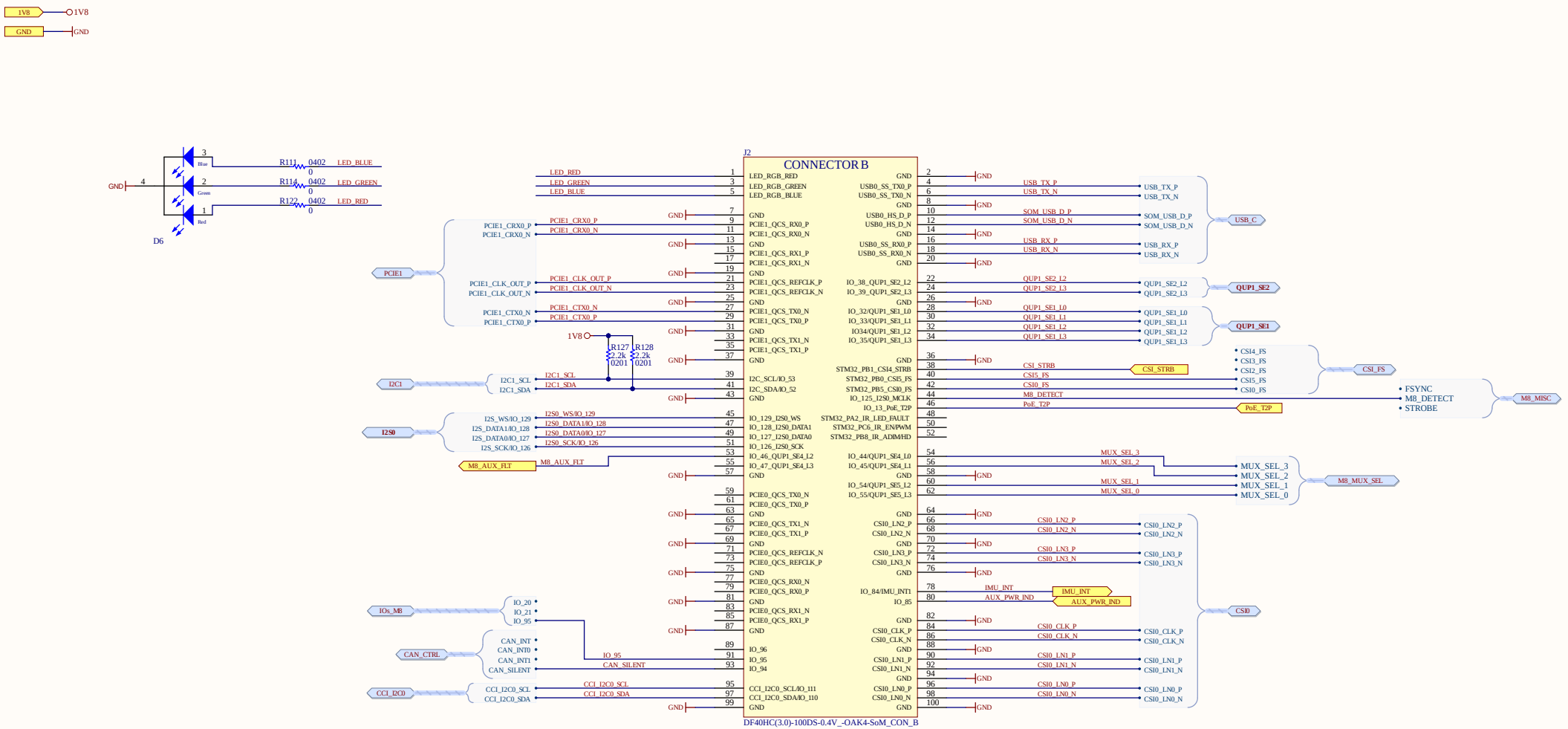
Luxonis



Title	BK3498		
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 9 of 21	
Drawn by: Blaz Kvas			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

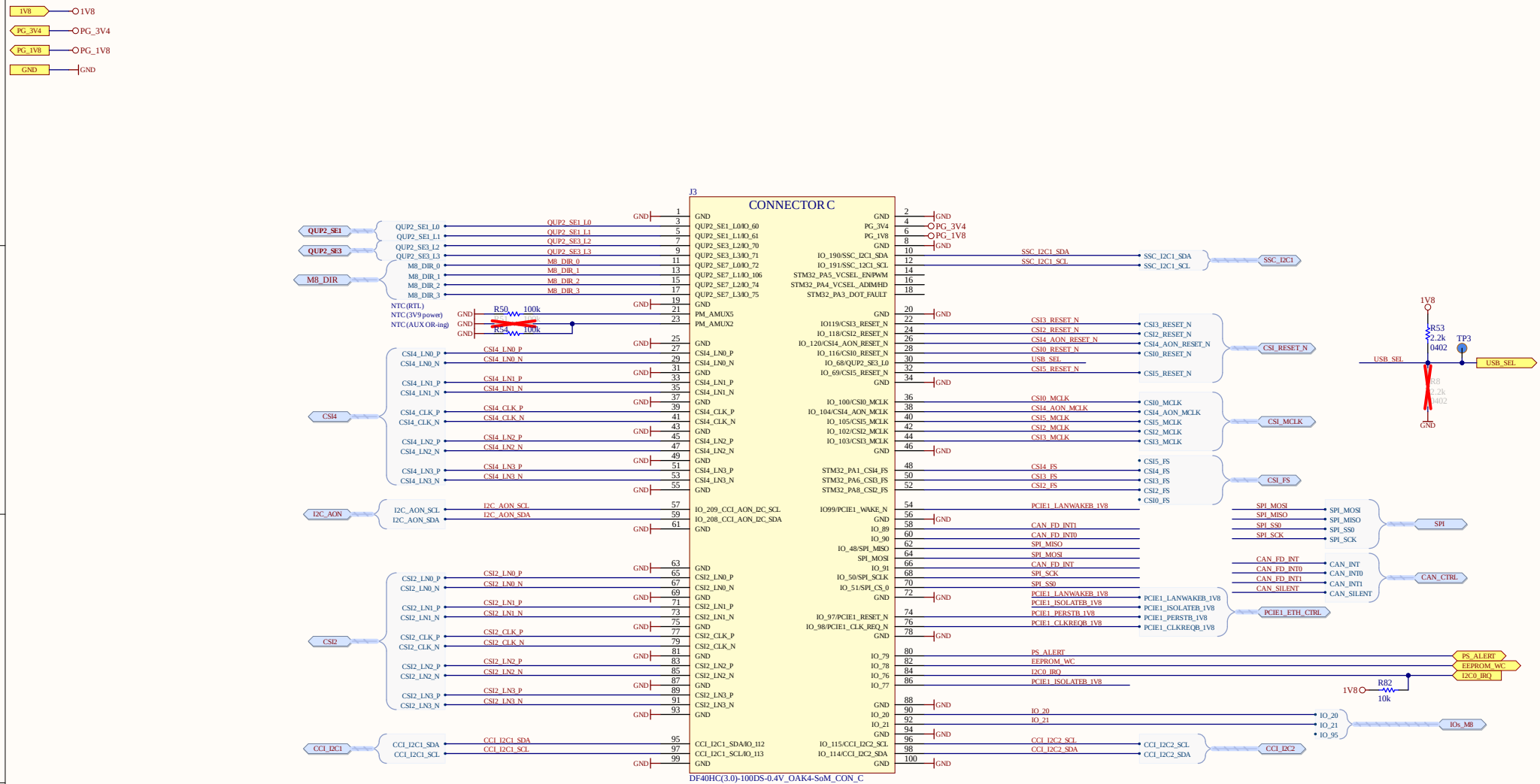
Luxonis



Title **BK3498**			
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 10 of 21	
Drawn by: *Blaz Kvas*			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

Luxonis



Title	BK3498		
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:43	Sheet 11 of 21	
Drawn by: Blaz Kvas			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

Luxonis

Operating temperature: 125°C (max)
Storage temperature: -55°C to 125°C
Ambient temperature: 0°C to 70°C
Junction-to-ambient: 19.06°C/W
Junction-to-top: 0.28°C/W

3V3 Supply Voltage: 3.14V (min), 3.46V (max)
3V3 Supply Current: 88mA (typ), 100mA (max)
0V9 Supply Voltage: 0.92V (min), 0.98V (max)
0V9 Supply Current: 364mA (typ), 650mA (max)

3V3
1V8
ETH_0V95
GND

- Place 100nF capacitors next to pin 29 and 46
- Place 2.2uF capacitors next to pin 19 and 33
- Place 10uF capacitor next to pin 19 (surge)
- Place 100nF capacitor next to pin 15 (surge)
- Place 10uF capacitors next to pin 8 and 28
- Place 2.2uF capacitor next to pin 47
- Place 2.2uF capacitors next to pin 11, 16 and 22
- Place 1uF capacitor next to pin 37

U10 RTL8125BI-CG

ETH_DATA

U11 AT93C46DY6-YH-E

U1 TXS0104ERUTR internal 10k PU

Title **BK3498**

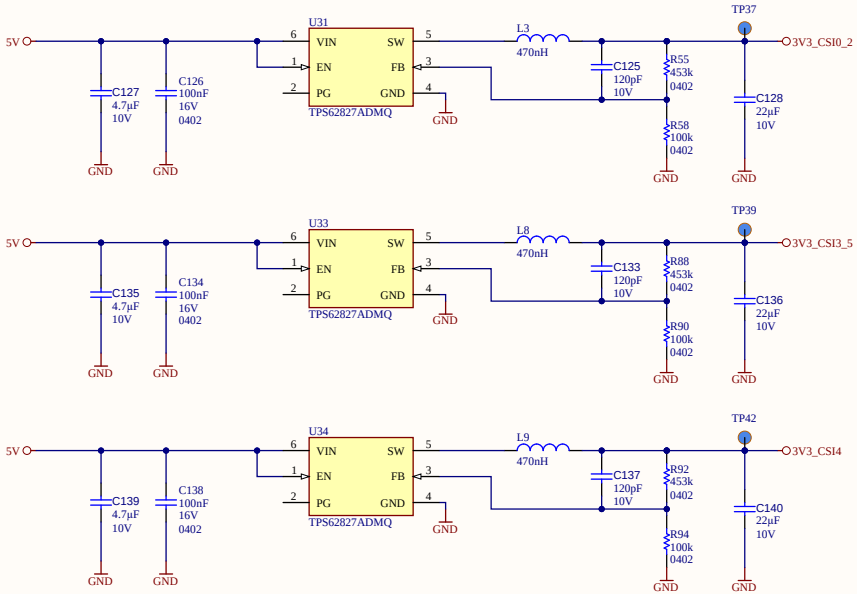
Size: Tabloid	Number: D2088000	Revision:
Date: 21/10/2026	Time: 11:51:44	Sheet 12 of 21

Drawn by: *Blaz Kvas*

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

Luxonis





Title **BK3498**			*Luxonis Holding Corporation* *1942 Broadway St.* *STE 314C,* *Boulder, CO 80302* *United States*
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 13 of 21	
Drawn by: ***Blaz Kvas***			

3V3_CSI4 → 3V3_CSI4

GND → GND

CSI_RESET_N: CSI3_RESET_N, CSI2_RESET_N, CSI4_AON_RESET_N → CSI4_AON_RESET_N, CSI0_RESET_N, CSI5_RESET_N

CSI_MCLK: CSI0_MCLK, CSI4_AON_MCLK → CSI4_AON_MCLK, CSI5_MCLK, CSI2_MCLK, CSI3_MCLK

CSI_FS: CSI5_FS, CSI4_FS → CSI4_FS, CSI3_FS, CSI2_FS, CSI0_FS

CENTER CAMERA CSI4

J10 — F32R-1A7H1-11026

Pin	Net	Function
MP1	GND	
26	3V3_CSI4	3V3
25		3V3
24	CSI_STRB	STROBE
23	CSI4_FS	FSIN
22	GND	GND
21	I2C_AON_SDA	SDA
20	I2C_AON_SCL	SCL
19	GND	GND
18	CSI4_AON_MCLK	CLK
17	CSI4_AON_RESET_N	PWR
16	GND	GND
15	CSI4_LN3_P	D3_P
14	CSI4_LN3_N	D3_N
13	GND	GND
12	CSI4_LN2_P	D2_P
11	CSI4_LN2_N	D2_N
10	GND	GND
9	CSI4_CLK_P	C_P
8	CSI4_CLK_N	C_N
7	GND	GND
6	CSI4_LN1_P	D1_P
5	CSI4_LN1_N	D1_N
4	GND	GND
3	CSI4_LN0_P	D0_P
2	CSI4_LN0_N	D0_N
1	GND	GND
MP2	GND	

BOTTOM CONTACT

I2C_AON → I2C_AON_SDA, I2C_AON_SCL

CSI4 → CSI4_LN3_P/N, CSI4_LN2_P/N, CSI4_CLK_P/N, CSI4_LN1_P/N, CSI4_LN0_P/N

D7 — TPD4E02B04DQAR

Pin	Net	Function
1	CSI4_AON_MCLK	IO1
2	CSI4_AON_RESET_N	IO2
3	GND	GND
4	CSI4_LN3_P	IO3
5	CSI4_LN3_N	IO4
10	CSI4_AON_MCLK	NC
9	CSI4_AON_RESET_N	NC
8	GND	GND
7	CSI4_LN3_P	NC
6	CSI4_LN3_N	NC

D8 — TPD4E02B04DQAR

Pin	Net	Function
1	CSI4_LN1_P	IO1
2	CSI4_LN1_N	IO2
3	GND	GND
4	CSI4_LN0_P	IO3
5	CSI4_LN0_N	IO4
10	CSI4_LN1_P	NC
9	CSI4_LN1_N	NC
8	GND	GND
7	CSI4_LN0_P	NC
6	CSI4_LN0_N	NC

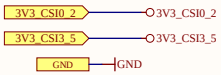
D19 — TPD4E02B04DQAR

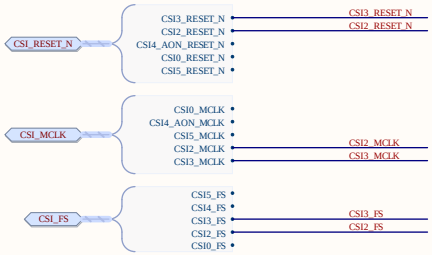
Pin	Net	Function
1	CSI4_LN2_P	IO1
2	CSI4_LN2_N	IO2
3	GND	GND
4	CSI4_CLK_P	IO3
5	CSI4_CLK_N	IO4
10	CSI4_LN2_P	NC
9	CSI4_LN2_N	NC
8	GND	GND
7	CSI4_CLK_P	NC
6	CSI4_CLK_N	NC

Title	BK3498		
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 14 of 21	
Drawn by: Blaz Kvas			

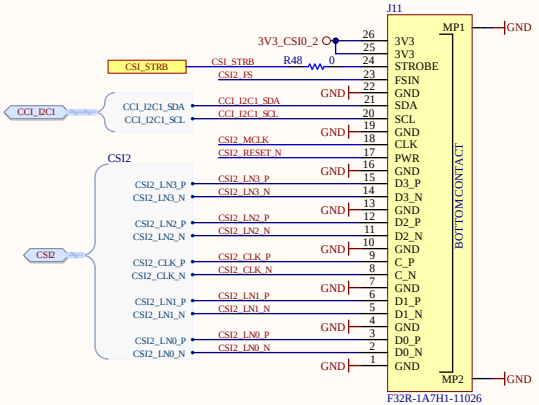
Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

Luxonis

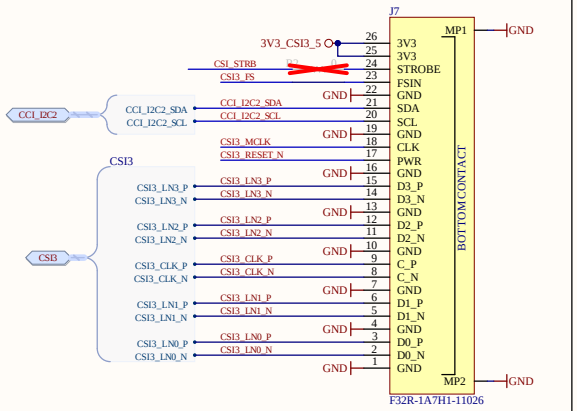


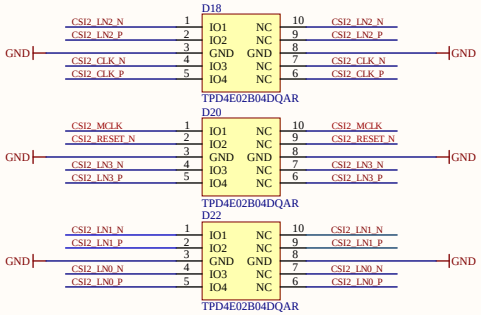


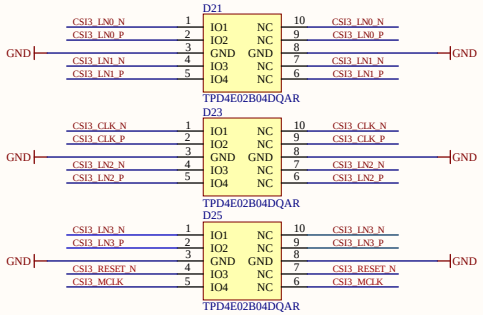
STEREO CAMERA CSI2



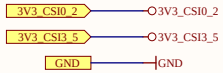
STEREO CAMERA CSI3

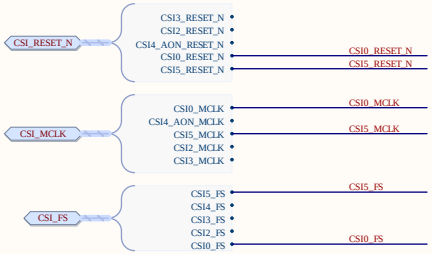


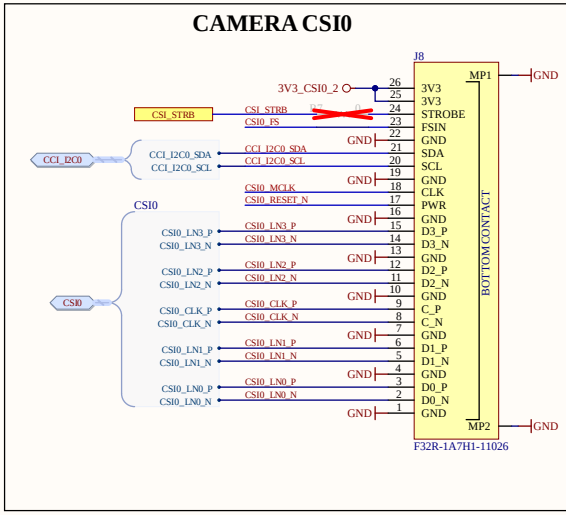


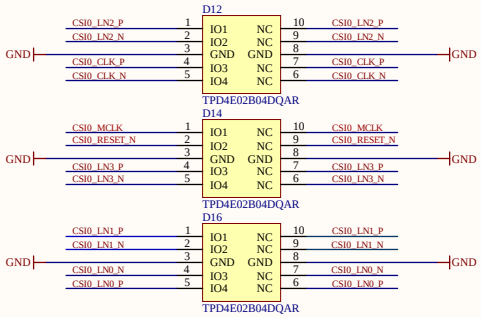


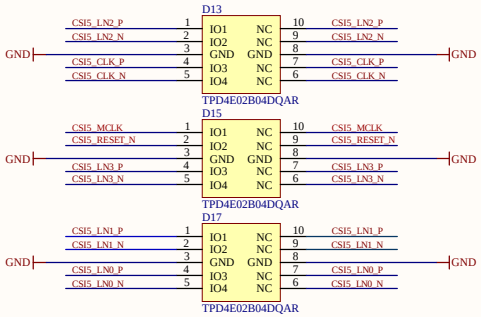
Title **BK3498**			*Luxonis Holding Corporation* *1942 Broadway St.* *STE 314C,* *Boulder, CO 80302* *United States*
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 15 of 21	
Drawn by: ***Blaz Kvas***			





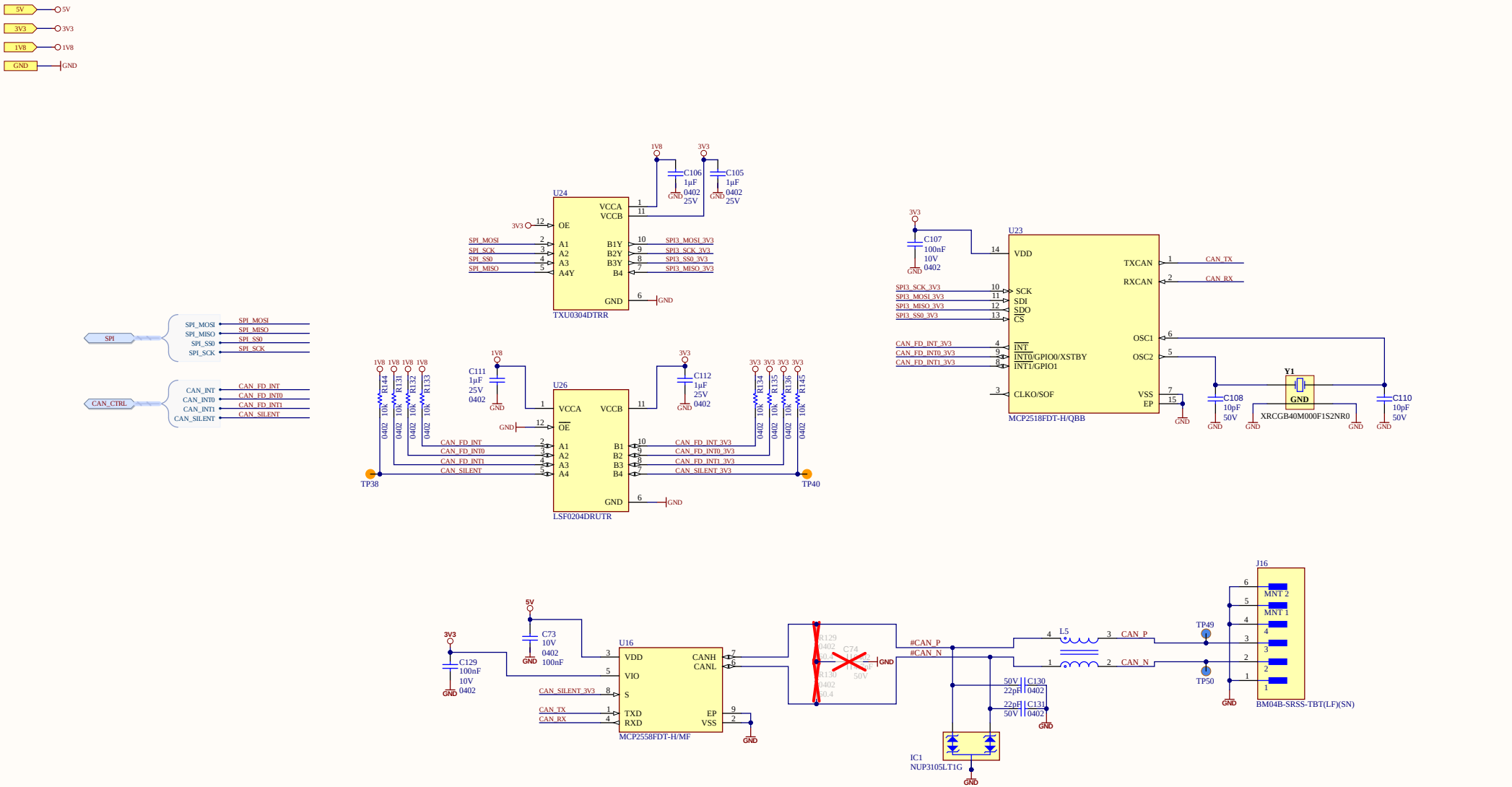






Title *			
Size: Tabloid	Number:*	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 16 of 21	
Drawn by: *			

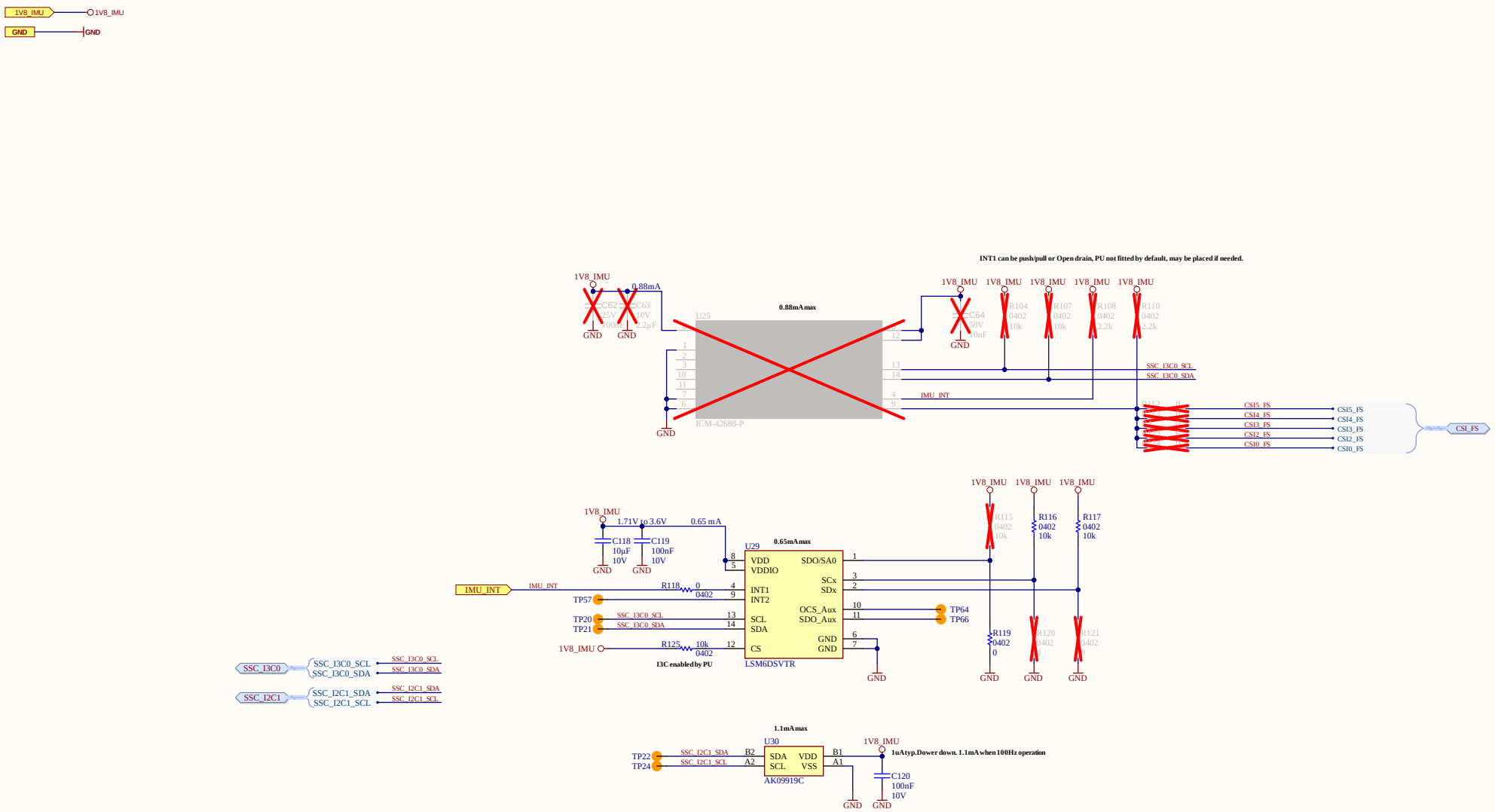
Luxonis



Title	**BK3498**		
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 17 of 21	
Drawn by: *Blaz Kvas*			

Luxonis Holding Corporation
1942 Broadway St.
STE 314C,
Boulder, CO 80302
United States

Luxonis

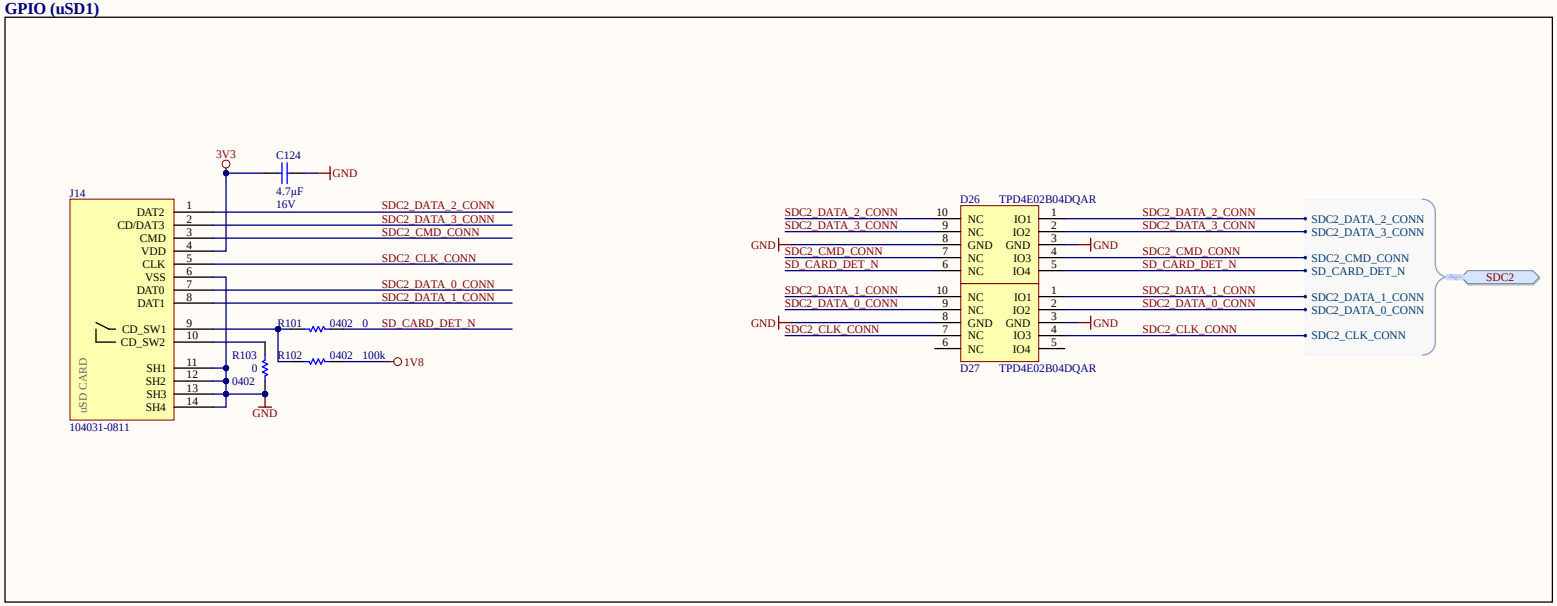


Title *			
Size: Tabloid	Number:*	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 18 of 21	
Drawn by: *			

Luxonis

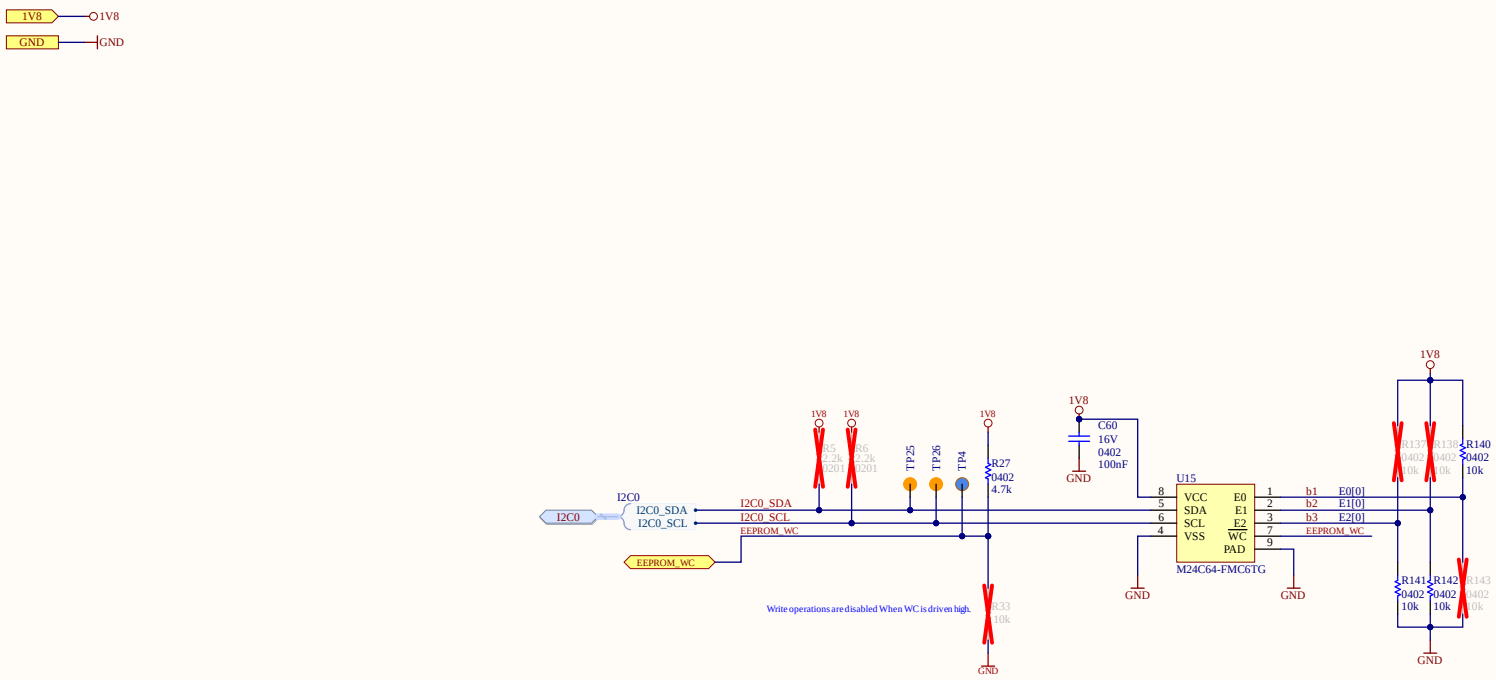
3V3
1V8
GND

GPIO (uSD1)

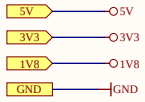


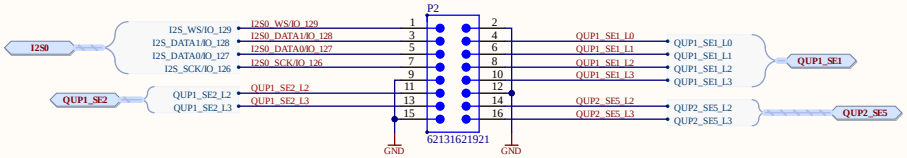
Title *			
Size: Tabloid	Number:*	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 19 of 21	
Drawn by: *			

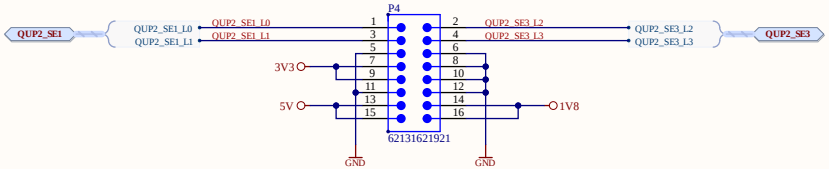
Luxonis



Title **BK3498**			*Luxonis Holding Corporation* *1942 Broadway St.* *STE 314C,* *Boulder, CO 80302* *United States*
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 20 of 21	
Drawn by: ***Blaz Kvas***			







Title **BK3498**			*Luxonis Holding Corporation* *1942 Broadway St.* *STE 314C,* *Boulder, CO 80302* *United States*
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:51:44	Sheet 21 of 21	
Drawn by: ***Blaz Kvas***			